

EDUCATION AND SCHOLASTIC ACHIEVEMENT			
Education	Institute	% / CGPA	Year
M Tech, Clinical Engineering	Indian Institute of Technology Madras, Tamil Nadu	7.63 CGPA	2026
B E, Biotechnology	Nitte Mahalinga Adyantaya Memorial Institute of Technology	8.39 CGPA	2022
Class XII (State)	Karkala Jnanasudha PU College, Udupi , Karnataka	93.66%	2018
Class X (State)	Swasthi Shree Nemisagara Varneeji, High school, Udupi, Karnataka	96.16%	2016
• Secured AIR 696 (All India Rank) in GATE 2024 – Biotechnology (BT) out of 17078 students.			
KEY PROJECTS			
Personal Project	Diabetic Readmission Prediction for High-Sensitivity Cleaning	(Jul-Aug 2025)	
• Optimized LightGBM for imbalanced diabetes readmission (89:11) using class weighting and threshold for minority class prediction.			
• Tuned key hyperparameters-learning rate, depth, leaves and regularization and evaluated using precision, recall, F1-score .			
• Achieved ~97% recall by selecting a custom decision threshold, balancing the recall–precision trade-off for high-sensitivity screening .			
Personal Project	Python Based Banking Management System	(Jan-Feb 2025)	
• Built a Python-based Banking Management System using OOP and Tkinter GUI for account management, transactions, and reporting.			
• Designed a modular structure with separate files for core functions, all linked to a single SQLite3 database for persistent storage.			
• Added admin authentication, input validation, and exception handling for secure, reliable, and real-time banking operations.			
Personal Project	ECG Classification using Autoencoder	(Mar-Apr 25)	
• Pre-processed MIT-BIH ECG beats and normalized them to a uniform scale, ensuring consistent input for model training.			
• Built a 1D convolutional autoencoder to compress ECG beats into 64 features with low reconstruction error (MSE ≈ 4e-5) .			
• Classified Normal vs. Abnormal ECG beats using latent features with Random Forest, achieving 97.8% accuracy and 0.993 ROC-AUC .			
Course-AM5510-IITM	Modelling And Its Evaluation For Time Series Data Using (AR, MA, ARMA, ARIMA) Models	(Sept-Nov 24)	
• Modelled and evaluated time series data like ECG and EMG using different models like Auto Regression, Moving Average, ARMA, ARIMA			
• Analysed the data using time domain, frequency domain methods. Used NumPy, Pandas & Matplotlib to implement those methods.			
Personal Project	MNIST Handwritten Digit Classifier	(Apr-Jun 25)	
• Built a CNN architecture with two convolutional layers, max pooling, dropout, and Xavier weight initialization for digit classification.			
• Trained the model using Adam optimizer, learning rate scheduler, and Cross-Entropy Loss with fully reproducible experimental settings.			
• Achieved 89.1% test accuracy with per-class precision, recall, and F1-score consistently above 91% , showing strong model performance.			
• Visualized training curves, sample predictions, and detailed performance metrics to thoroughly evaluate and interpret model behaviour.			
Personal Project	House Price Prediction Using Randomforestregressor	(Jan-Feb 25)	
• Pre-processed Beijing housing dataset by handling missing values, outliers, and data inconsistencies, ensuring reliable model training.			
• Developed and trained a RandomForestRegressor with 200 trees , effectively capturing complex non-linear relationships of features.			
• Achieved Validation/Test RMSE ≈ 0.04 and R² ≈ 0.998 , demonstrating the model's ability to generalize well than baseline model.			
Course-SCTIMST*	Brain Tumor classification	(Aug 25- present)	
• Developing a deep learning model for brain tumor classification by MRI images with normalization, resizing, and augmentations.			
• Conducting training and validation using optimized architectures, loss functions, and learning schedules to enhance performance.			
• Evaluating model predictions, generating performance metrics and iteratively improving the system for reliable tumor detection.			
Personal Project	Automated Pneumonia Classification from Chest Radiographs Using DenseNet201	(May-Aug 25)	
• Developed a DenseNet201-based TensorFlow/Keras model for pneumonia detection using Kaggle chest-Xray pneumonia dataset.			
• Applied preprocessing with RGB conversion, normalization, and augmentation to X-ray images, optimized with GPU in Google Colab .			
• Achieved ~90% training and ~89% test accuracy for accurately classifying normal vs. pneumonia cases.			
Course-CMC* Vellore	COHMET - Cognitive and Hand Motor skill Enhancement Trainer	(Mar-Jun 25)	
• Designed & developed a smart electronic pegboard for cognitive and motor rehabilitation of stroke and spinal cord injury patients .			
• Designed mechanical structures for sensor integration & placement of addressable LEDs using SketchUp and fabricated using 3d printer .			
• Built custom app to control the device via HC05 module and store the patient data to google sheets API accessible via custom website .			
• Successfully tested the device on 5 different patients at CMC* Vellore rehabilitation centre with validation from doctors and therapists .			
Relevant Courses, Tools, and Skills			
• Linear Algebra • Biostatistics • Computer Aided Design • Basic Programming • Numerical Methods • Heat and Mass Transfer • Applied Biostatistics • Bioprocess principles and calculations	• Virtual Reality Engineering • Biomedical Signal And Systems • Biomedical Sensors and Measurement • Biomechanics • Biomedical Imaging • Clinical Attachment	• Quality system management • Biomaterials • Design Tools for Clinical Engineering • Medical Device Technology • Critical Care Instrumentation • Physiology and Anatomy • Unit operations in bioprocess industries	▪ Language -Python ▪ Software - Creo, ANSY, VS code, Figma, Sketchup, Arduino IDE, Git, GitHub, Unity, Jupyter notebook, Google Colab ▪ Skills - Web Dev (basic), GUI, OOPS, 3D Modelling, Graphics Design, Prototyping, 3D printing & product development, documentation, ▪ Framework - Tkinter, NumPy, Pandas, Matplotlib, OpenCv, SQLite, Flask, TensorFlow, Keras. Google Cloud API, MediaPipe
Hobbies and Others			
Hobbies -Bharatnatyam, Hiking, Cooking, Yoga, Swimming, reading books. Languages - English, Hindi, Tulu, Kannada			